

TITLE: Outdoor Small Uncrewed Aircraft System

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Public Safety Research Division (671.00)

General Statement of Need:

The Public Safety Research Division (PSCR) is in need of an outdoor-flight capable small Uncrewed Aircraft System (sUAS) for flight testing comparison of drone simulator and to develop standard test methods for link loss research. This aircraft will be used for laboratory demonstrations as well as scientific research of first responder UAS. PSCR desires to purchase a specific sUAS that meets the required specifications.

Item Specifications and Quantities:

Item 1:	Small Uncrewed Aircraft System Airframe & Operator Control Unit	
Quantity:	1 EACH	
Specifications:	1.1	Modular sub-55-pound multirotor sUAS capable of outdoor flight
	1.2	NDAA and/or TAA Compliant
	1.3	Durable construction, capable of surviving crashes with minimal repair, such as propeller replacement
	1.4	2.4/5 GHz, 5G/LTE ready (Operator Control Unit and Aircraft)
	1.5	IP55 (dust & water resistant) - Aircraft
		IP54 – Operator Control Unit
		Operator Control Unit with integrated touch screen and stick controls
	1.6	Sub-60 startup time
	1.7	Flight time minimum of 40 minutes
	1.8	Operational in GPS-denied position-hold
	1.9	Autonomy including true 360-degree obstacle avoidance coverage
	1.10	Onboard vision system with lights
	1.11	4K minimum video resolution
	1.12	Infrared and telephoto navigation cameras
	1.13	Gimbal stabilization with light modes (~1000 lumens)
	1.14	Two 256 GB micro-SD cards
	1.16	Three batteries and one dual charger
	1.17	Ruggedized storage case, custom fit for the sUAS, controller and additional batteries
	1.18	One set of spare propellers
	1.19	Windspeed handling (gust) of no less than 11.17 m/s (25 mph)
Item 2:	Attachments	
Quantity:	1 SET	
Specifications:	2.1	RTK/PPK GPS
	2.2	Payload Dropper
	2.3	Parachute
	2.4	Spotlight
	2.5	Speaker

Vendor Requirements

1. Possess at time of award/contracting an Export Control Classification Number (ECCN).
2. Possess at time of award/contracting a Unique Entity Identification Number (UEI) which registration is listed as “Active” on the GSA System for Award Management (SAM.gov).

Procurement Support Information and Ancillary Services:

[Provide input for each section Below or Mark Not Applicable (N/A)]

Section	Requirements/Specifications
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.] ☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.) ☐ Vendor Representative On-Site more than three (3) days. ☐ Other Considerations: [List other considerations to ensure an accurate quote.]
Shipping &	Shipping Address:

Delivery ☐ N/A	☒ NIST-Boulder ATTN: Provided at award Mail Stop: 325 Broadway Boulder, CO 80305 ☐ NIST-Gaithersburg ATTN: Mail Stop: 100 Bureau Dr Gaithersburg MD 20899 ☐ [Alternate Shipping Address] Shipping Criteria: ☒ Partial Delivery Acceptable ☐ [Enter Shipping and Delivery information by item.] ☐ Direct Delivery to OU Building Required [Explain Need for Requirement] Building /Room Number: [Enter Building/Room Number] ☐ Building has Loading Dock Containerization Preference: [Enter Container Type] Other Requirements: [List Any Other Requirement] Delivery Date Criteria: ☒ Delivery Shall Be Completed No Later Than: 19 AUG 26 (08/19/2026) ☐ Delivery Shall Be Made No Earlier Than: Click or tap to enter a date. ☐ Scheduled Deliveries: <table><tr><th>Item Number</th><th>Due Date</th></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table> ☐ Other: [List Other Shipping, Delivery or Special Requirements.]	Item Number	Due Date				
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Inspection and Acceptance ☐ N/A	☐ On-Site ☒ Acceptance Not Expected to Exceed 7 Business Days ☐ Acceptance Expected to Exceed 7 Business Days: [Duration] ☐ Inspection and Acceptance Plan: [Explanation] ☐ Other: [Enter Other Requirements]
Training ☒ N/A	Training Requirements (Who, When, What, Where) ☐ Number of Trainees: [Enter Number of Trainees] ☐ On-Site ☐ Off-Site Location: [Enter Location(s)] Is Trainee Travel Required: [Choose Answer] Scope of Training ☐ Operation☐ Maintenance☐ Troubleshooting ☐ Safety☐ Other [Enter Other Scope of Training Requirements] ☐ Training Materials to be Provided by Government: [Describe Materials] ☐ Training Materials to be Provided by Vendor: [Describe Materials] ☐ Other [Enter Other Training Requirements] NOTE: SME shall be coordination POC.
Warranty ☐ N/A	Base Manufacturer Warranty is acceptable Warranty Length, Base Period: 1-Year What is included? ☒ Labor☒ Parts☐ Travel ☐ Cost and Liability for Returns ☒ Other One full aircraft replacement ☐ 1 Option Period ☒ 2 Option Periods of 1-Year Each ☐ 3 Option Periods ☐ 4 Option Periods (Maximum) Total Warranty Length: 3-Years
Calibration ☒ N/A	[Describe Calibration Requirements Per Item] Calibration Plan Length: [i.e. 3 - Years] What is Included? ☐ Labor☐ Spare Parts☐ Consumables ☐ Travel☐ Cost and Liability for Shipment ☐ Software Updates☐ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Maintenance ☐ N/A	[Describe Maintenance Requirements Per Item] Maintenance Plan Length: 1-Year What is Included? ☐ Labor☒ Spare Parts☐ Consumables ☐ Travel☐ Cost and Liability for Shipment ☒ Software Updates☒ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period ☒ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Government	List Any Government Furnished Material / Equipment Per Item]

Requirements / Specification Document

Furnished Material / Equipment ☒ N/A	☐ Samples for Testing ☐ Equipment which will leave Government Site Description: [Enter Description] Serial Number: [Enter Serial Number] NIST Property Number: [Enter NIST Property Number] ☐ Other: [Enter Other Details]
Travel ☒ N/A	[Describe any Travel Requirements Per Item] ☐ Explain: [Enter Justification / Requirement] ☐ Other: [Enter Other Requirements] NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.